Seminar 04 cheat sheet

# Accelerated Gradient Methods

**Optimization Methods**
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## SPD quadratic and GD

- $f = \frac{1}{2}x^\top Qx - b^\top x$, $x^* = Q^{-1}b$, $\kappa = L/\mu$.
- $0 < \alpha < 2/L$; best fixed quadratic step $2/(L+\mu)$.

## Heavy ball

- $x^+ = x - \alpha g(x) + \beta(x - x^-)$; $x_{-1} = x_0$.
- $p_\lambda(r) = r^2 - (1 + \beta - \alpha\lambda)r + \beta$.
- For $0 \le \beta < 1$: $0 < \alpha < 2(1+\beta)/L$.
- $\alpha_* = \frac{4}{(\sqrt{L}+\sqrt{\mu})^2}$, $\beta_* = (\frac{\sqrt{L}-\sqrt{\mu}}{\sqrt{L}+\sqrt{\mu}})^2$.
- $\rho_* = (\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)$. Quadratic result; transient growth is possible.

## Look-ahead

- $y = x + \beta(x - x^-)$, $x^+ = y - \alpha g(y)$.
- Gradient at the extrapolated point. A restart discards momentum and retries.

## Coordinate descent

- $L_i = Q_{ii}$, $x_i^+ = x_i - g_i/L_i$. Decrease $g_i^2/(2L_i)$.
- Greedy: maximize $|g_i|/\sqrt{L_i}$. Cyclic and uniform random are different selectors.
- $d$ updates = one pass. Count the cost of a greedy scan.

## CG for an SPD system

- $r = b - Ax$, $p = r$ initially. Reuse $q = Ap$.
- $\alpha = \frac{r^\top r}{p^\top q}$; $x^+ = x + \alpha p$, $r^+ = r - \alpha q$.
- $\beta = \frac{(r^+)^\top r^+}{r^\top r}$; $p^+ = r^+ + \beta p$.
- $r_i^\top r_j = 0$, $p_i^\top A p_j = 0$ in exact arithmetic, $i \ne j$.
- Check true $b - Ax$; relative residual uses $\|b\|$ if nonzero.

## SGD and variance

- $F = N^{-1}\sum_i \ell_i$, $g_B = |B|^{-1}\sum_{i\in B}\nabla\ell_i$.
- Independent draws: $\mathrm{Var}(g_B) = \sigma^2/B$.
- Scalar quadratic: $V_\infty = \frac{\alpha\sigma^2}{B\mu(2-\alpha\mu)}$, $0 < \alpha\mu < 2$.
- Shuffle each epoch; include and correctly normalize the last short batch.

## Schedules and work

- $\alpha_0$, $\alpha_0/\sqrt{1+k}$, $\alpha_0/(1+k)$; $k$ counts updates.
- Compare 5 seeds, the central curve and the quartile interval. Count sample gradients and full-loss logging.

## Feature scaling

- $Z = (X - m)S^{-1}$, $H_Z = S^{-1}HS^{-1}$.
- Original coefficients $w = S^{-1}u$, intercept $b = c - m^\top w$.
- Use training statistics. Scaling does not remove correlation.

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